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FULL LENGTH ARTICLE

Identification of aircraft longitudinal aerodynamic parameters using an online corrective test for wind tunnel virtual flight



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Abstract The identification of aerodynamic parameters is accomplished through the test data of the dynamic movement of scaled aircraft models flying dynamically in wind tunnel, which can realize the accurate acquisition of the aerodynamic model of the aircraft in the preliminary stage for aircraft design, and it is of great significance for improving the efficiency of aircraft design. However, the translational motion of the test model in the wind tunnel virtual flight is subject to constraints that result in distinct flight dynamics compared to free flight. These constraints have implications for the accuracy of aerodynamic derivatives obtained through the identification of wind tunnel test data. With this issue in mind, the research studies the differences in longitudinal dynamic characteristics between unconstrained free flight and wind tunnel virtual flight, and innovatively proposes an online correction test based wind tunnel virtual flight test technique. The longitudinal trajectory and velocity changes of the model are solved online by the aerodynamic forces measured during the test, and then the coupled relationship between aircraft translation and rotation is used to correct the model's pitch attitude motion online. For the first time, the problem of solving the data approximation for free flight has been solved, eliminating the difference between the dynamics of wind tunnel virtual flight and free flight, and improving the accuracy of the aerodynamic derivative identification results. The experiment's findings show that accurate aerodynamic

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derivatives can be identified based on the online correction test data, and the observed behaviour of the identified motion model has similarities to that of the free flight motion model.

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1. Introduction

Accurate aerodynamic model is important for the overall program design of the aircraft as well as the design of the flight control laws. In general, there are two types of tests for obtaining an aerodynamic model of aircraft: force measurement tests in wind tunnels and aerodynamic parameter identification based on flight test data. Conventional wind tunnel tests have disadvantages such as interference from fixed mounts, interference from rotating balance mounts, and the need for separate measurements of static and dynamic derivatives.¹ Aerodynamic derivatives are often identified using real aircraft flight test data, and although this approach yields reliable findings, there are a number of drawbacks, including high risk and high expense.^{2,3} Between the conventional wind tunnel test and the actual flight test derived from a new type of wind tunnel test technique, in the predesign phase of the aircraft, through the scaled aircraft model in the wind tunnel to replicate dynamic motion and authentic flying conditions, and after that finish identifying the aerodynamic derivatives, this innovative testing method is known as the Wind Tunnel Virtual Flight Test (WTVFT).^{4,5} As seen in Fig. 1, this paper carries out research on a 3-DOF WTVFT. By connecting the test model to the strut via a three-axis rotation mechanism with the model's centre of mass at the point of connection, the model maintains a fixed position while possessing degrees of freedom along three axes of rotation.^{6,7} By adjusting the servo's deflection, the test model's dynamic motion in the wind tunnel is achieved, which is the difference from the conventional wind tunnel experiment, i.e., the model can move autonomously instead of being forced to oscillate.⁸

However, the results obtained from aerodynamic parameter identification by means of test data from conventional WTVFT can differ from the theoretical values. In the WTVFT, some of the motion degrees of freedom in the model are constrained, leading to differences in its dynamics from actual free flight, the model's motion reactions as determined by the test are not the same as those from real flight.⁹ There-

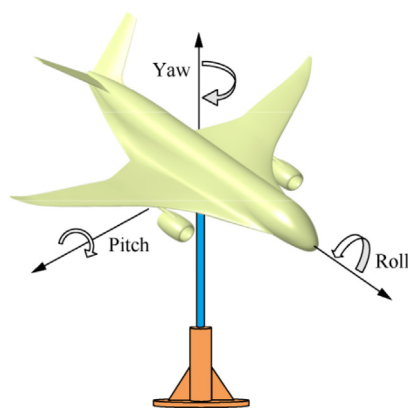


Fig. 1 A paradigm for doing wind tunnel virtual flight experiments with 3-DOF.

fore, if the WTVFT data are directly utilized for aerodynamic parameter identification, the results obtained are not accurate.

To address the above problems, it is necessary to improve the accuracy of the identification results by analysing the difference between WTVFT and free flight and proposing a correction method for wind tunnel tests.¹⁰ At present, numerous investigations based on WTVFT have been conducted to identify aerodynamic parameters, all of which are aimed at improving the accuracy of the identification results.

These studies can be summarized in two aspects, namely, increasing the degrees of freedom of motion of the wind tunnel test setup and carrying out online correction tests for the WTVFT.

- (1) The difference in flight dynamics between WTVFT and free flight can be reduced by increasing the motion degrees of freedom of the test model. A 5-DOF WTVFT was conducted at the University of Bristol, UK, using a BAe Hawk aircraft, i.e., three-axis rotation of the model around the centre of mass superimposed on the pitch and yaw rotational motions of the support arm.¹¹ However, there is a force and moment interaction between the test model and the support arm, which leads to the interference of the model's motion by the rotational motion of the support arm and thus will have an impact on how accurate the identification results are.^{12,13} Cranfield University also conducted a 4-DOF WTVFT using a BAe Hawk aircraft, adding vertical lift and sink motion to the three-axis rotation.^{14,15} But because of the wind tunnel's size limitations, the range of the biggest heave was about 0.75 m, which could not adequately present the model's longitudinal motion modes during the test,¹⁶ therefore influencing the identification findings' accuracy. The two approaches listed above are used to reduce the distinction between real free flight and the WTVFT by increasing the motion degrees of freedom of the test model, but the issue with both is that they have an impact on the identification findings' accuracy due to the restricted motion range of the test model.
- (2) The U.S. Air Force Research Laboratory has proposed the concept of an online correction test, i.e., the test model's attitude change is measured by a WTVFT, while the trajectory and velocity changes are obtained by online mathematical simulation. The identification of aerodynamic parameters is then done using the corrected test results, but this idea has not been verified by theoretical analysis and wind tunnel testing.^{17,18}

To guarantee that the aerodynamic derivatives can be identified, it is necessary to design the excitation signal's amplitude and frequency range. There are two existing methods for the design of excitation signals: one is to design optimal signals by ensuring that different paradigms of the information matrix (which can also be the covariance matrix of the observation

error) take extreme values or to guarantee that any aerodynamic derivative can be identified by designing complex orthogonal multisinusoidal optimized excitation signals.^{19,20} However, both of those approaches are laborious and require a lot of computing power.

In light of the aforementioned difficulties, this paper proposes a technique for identifying an aircraft's longitudinal aerodynamic parameters according to an online correction test of WTVFT. First, the variations in the aircraft's longitudinal dynamics during WTVFT and free flight is analysed, and the online correction test and data solution method for WTVFT are established, so as to eliminate the difference between the flight dynamic characteristics of WTVFT and free flight. Second, an aerodynamic parameter identification model applicable to the corrected test data is developed. In addition, it is suggested to use a frequency domain analysis-based approach to design the excitation signal's parameters by doing analysis of Bode diagrams on the modified identification model. Finally, the identification of longitudinal aerodynamic derivatives is completed using corrected test data. By comparing and analysing the identification results of the corrected and uncorrected test data, the process of validating this paper's methodology has finished. The following is a summary of this paper's contributions:

- (1) A new approach to online correction tests for WTVFT is proposed. The longitudinal trajectory and velocity changes of the model are solved online using the measured aerodynamic forces. The model's pitch attitude motion is corrected online based on the coupling relationship between the aircraft's translational and rotational motions, and then the solved motion variables are fed back to the flight control law of the test model. The differences in flight dynamics between WTVFT and free flight are eventually eliminated.
- (2) An innovative approach to designing the amplitude magnitude and frequency band range of an excitation signal is put forward, which is based on frequency domain analysis. The excitation signals are designed to ensure that the response amplitude of the aerodynamic force or aerodynamic moment corresponding to each aerodynamic derivative is high enough to ensure the recognizability of all aerodynamic derivatives.
- (3) Compared with the aerodynamic derivatives obtained based on the conventional WTVFT data, the identification results obtained based on the online correction test data have less deviation from the theoretical values, and more accurate aerodynamic derivative identification results can be obtained.

Subsequent sections of this paper are structured as follows: Section 2 describes the methodology for online correction tests and data solving for WTVFT. The development of the identification model and the design of the excitation signal are described in Section 3. Section 4 verifies the correctness and rationality of the methodology of this paper, and Section 5 is the conclusion.

2. Online correction test and data solving for WTVFT

According to Fig. 1, in the WTVFT, the test model carries out free rotation motion in three axes around the centre of mass.

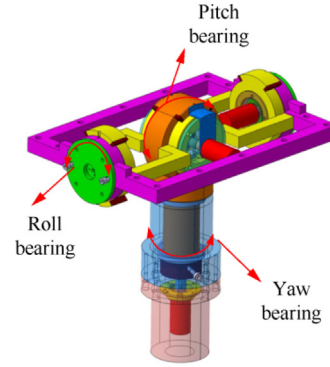


Fig. 2 Rotating device with a total of 3-DOF.

The mechanism of three-axis rotation consists of three deep groove ball bearings and rotation frames, as shown in Fig. 2, with rotation ranges of $\pm 45^\circ$, $\pm 180^\circ$, and $\pm 45^\circ$ for roll, yaw, and pitch, respectively.²¹

The pitching control surface's deflection propels the test model's longitudinal pitching motion in the wind tunnel. The model has on-board sensors attached that measure its motion variables; aerodynamic parameter identification is done using these test results. Furthermore, the model's centre of mass is fixed in the 3-DOF WTVFT, and a strain gage (Fig. 3) is affixed to the model's centre of mass in order to record the aerodynamic forces (lift, drag, and side forces) that are continuously applied to the model. Due to the low wind speed (about 30 m/s) and smooth airflow during the test, it can be approximated that the centre of mass of the model cannot move and the inertial force generated by the model can be neglected.²² Table 1 presents the data that can be obtained through direct measurement during the wind tunnel test.

The test yields measurement accuracy values of $\pm 0.1^\circ$ for the attitude angle, $\pm 0.1^\circ/\text{s}$ for the angular rate, and $\pm 0.1^\circ$ for the control surface deflection angle. The performances of the sensors used in the test are presented as Table 2, and throughout the test, the test results are captured at a 50 Hz sample frequency.

2.1. Differences in dynamics between WTVFT and free flight

The translational motion of the model is constrained by the support device, allowing only its rotational degree of freedom. As a result, the total forces acting on the model, including gravity, triaxial aerodynamic force, and the force exerted by the support device, sum up to zero, and the longitudinal dynamical equations for the model's centre of mass are shown in Eq. (1):²³

$$\begin{cases} F_x = m(\dot{u} + qw - rv) = 0 \\ F_z = m(\dot{w} + pv - qu) = 0 \end{cases} \quad (1)$$

where F_x and F_z represent the total external forces acting on the model in the body coordinate system, m is the model's mass, and u , v , and w are the airspeed components.

The dynamics and kinematic equations of the model's longitudinal rotation around its centre of mass, ignoring its lateral motion, are the same as those of free flight, as shown by Eq. (2):

$$\begin{cases} M = I_y \dot{q} \\ \dot{\theta} = q \end{cases} \quad (2)$$



Fig. 3 Strain gauge balance.

Table 1 Data obtained from simulated flight test conducted in wind tunnel.

Description	Instruments used for measurements
Roll angle, pitch angle, and yaw angle (ϕ , θ , ψ)	Inertial Measurement Unit (IMU)
Roll rate, pitch rate, and yaw rate (p , q , r)	IMU
Aerodynamic forces in the body coordinate system (X_b , Y_b , Z_b)	Strain gage balance
Angle of attack, sideslip (α , β)	Numerical solution
Elevator, rudder, and aileron deflections (δ_e , δ_r , δ_a)	Rotary encoder

Table 2 Sensor's performance metrics.

Sensors	Parameters	Performances	
		Bandwidth (Hz)	Measurement range
IMU	ϕ , θ , ψ	30	$\pm 180^\circ$
	p , q , r	30	$\pm 200^\circ/\text{s}$
Rotary encoder	δ_e , δ_r , δ_a	100	$\pm 180^\circ/\text{s}$

where I_y is the pitching moment of inertia for the model; and M is the pitching moment applied to the whole aircraft.

When identifying aerodynamic characteristics, the angle of attack and sideslip angle are crucial parameters whose measurement precision directly impacts the identification findings' precision. In the real flight, the airflow angle is usually measured by a wind vane sensor, which has a measuring inaccuracy because to the airflow instability. Nonetheless, during the WTVFT, the attitude angle can be used to directly solve for the sideslip and angle of attack. The solved airflow angle has a high accuracy since the attitude angle measurement accuracy is good as well.

The ground coordinate system is defined by the origin O_g , which is located at the model's centre of mass. $O_g x_g$ represents the direction of the wind speed, whereas $O_g z_g$ points vertically downwards. The incoming flow direction remains constant throughout the WTVFT, which causes the x -axis of the ground coordinate system and the airflow coordinate system to coincide. Eq. (3) illustrates how to solve the attitude angle and coordinate system transformation matrix to determine the airspeed component in the body coordinate system:

$$\begin{cases} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = L_{bg} \begin{bmatrix} V \\ 0 \\ 0 \end{bmatrix} \\ L_{bg} = \begin{bmatrix} \cos \theta \cos \psi & \cos \theta \sin \psi & -\sin \theta \\ \sin \phi \sin \theta \cos \psi - \cos \phi \sin \psi & \sin \phi \sin \theta \sin \psi + \cos \phi \cos \psi & \sin \phi \cos \psi \\ \cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi & \cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi & \cos \phi \cos \psi \end{bmatrix} \end{cases} \quad (3)$$

where the matrix used to translate the ground coordinate system into the body coordinate system is called L_{bg} . The expressions for the angle of attack α and the sideslip β are obtained from the airspeed component derived from Eq. (3), as shown in Eq. (4):

$$\begin{cases} \alpha = \tan^{-1} \left(\frac{w}{u} \right) = \tan^{-1} \left(\frac{\sin \phi \sin \psi + \sin \theta \cos \phi \cos \psi}{\cos \psi \cos \theta} \right) \\ \beta = \sin^{-1} \left(\frac{v}{V} \right) = \sin^{-1} (-\cos \phi \sin \psi + \sin \theta \sin \phi \cos \psi) \end{cases} \quad (4)$$

The model has no translational degrees of freedom during WTVFT, causing the motion characteristics of it and free flight to vary. The specific differences between the two need to be investigated, and the parameters of the motion that need to be solved and corrected online during the WTVFT need to be clarified.

The following factors account for the variations in longitudinal movement characteristics between WTVFT and unrestrained free flight:

- (1) During the WTVFT, the wind tunnel's wind speed is constant, and there is no longitudinal trajectory motion or velocity change for the test model. Therefore, the model's aerodynamic forces must be recorded using online correction tests and reconstruct the longitudinal trajectory and velocity changes of the model online based on the aircraft's translational equations of motion.
- (2) According to Eq. (5), for free flight, the angle of attack (α) and the climb angle (γ) add together to almost equal the pitch angle (θ) when lateral-directional motion is not considered.²⁴ During the WTVFT, the test model's climb angle γ is always 0° , so the angle of attack and pitch angle are approximately equal.

$$\begin{cases} \theta = \alpha + \gamma & \text{Free flight} \\ \theta = \alpha & \text{WTVFT} \end{cases} \quad (5)$$

Since the longitudinal translational motion and pitching motion are coupled with each other, for the test model in the WTVFT, there is no translational motion, which leads to the difference in pitch angle and angular velocity between the WTVFT and the free flight. Therefore, it is also essential to correct the response θ , q of the pitching motion for the test model online based on the coupling relationship between translational and rotational motion.

2.2. Methods for online correction tests

If the WTVFT identifies the model's longitudinal aerodynamic derivatives throughout the whole envelope range, it must be carefully adjusted to vary within the intended range in terms of pitch angle.¹⁰ Therefore, the test model uses the pitch angle command's control stabilisation control law structure, as shown in Fig. 4.

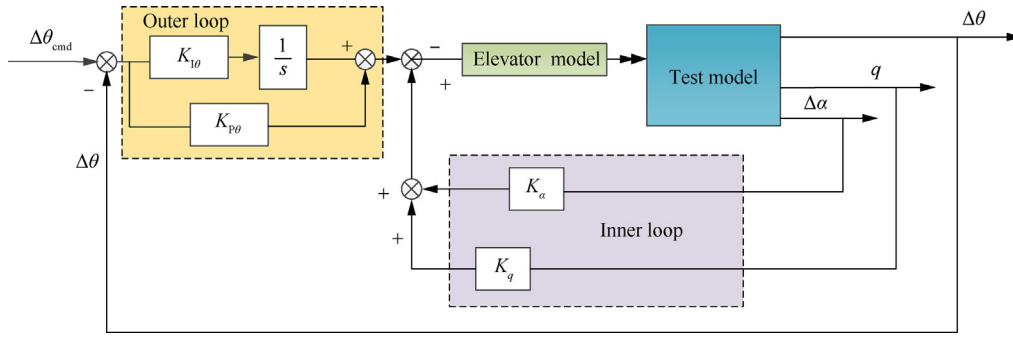


Fig. 4 Structure of longitudinal control law.

The parameters K_z , K_q in the inner loop of the control law are used to increase the longitudinal static stability and the damping of the pitch motion of the test model, while the parameters $K_{p\theta}$ and $K_{I\theta}$ in the outer loop of the control law are used to control the precise tracking of the test model with the pitch angle command θ_{cmd} .

According to the analysis in Section 2.1, there is no translational motion or velocity change in the model during the WTVFT, and the online correction test can realize the online solution and correction of the data used for identification so that the results can approximate the simulation of the real flight.

In free flight, the velocity direction of the longitudinal movement of the aircraft is determined by the climb angle γ . Variations in the climb angle result in variations in the flight altitude, and variations in the aircraft height influence variations in the airspeed magnitude. Therefore, to eliminate the differences caused by the displacement constraints as well as the constant velocity, it is essential to deal with the model's longitudinal trajectory and velocity variation online during the online correction test of the WTVFT and correct the pitching motion of the aircraft at the same time. The specific online correction test process is shown in Fig. 5.

First, as seen in Fig. 5, the force balance measures the model's three-axis aerodynamic forces (X_b , Y_b , and Z_b) within real time throughout the wind tunnel test, and these aerodynamic forces are transmitted to the simulation computer. The longitudinal positions x_{corr} and H_{corr} , climb angle γ_{corr} and airspeed variation V_{corr} of the model are solved by online simulation based on the aircraft's centre-of-mass translational equations of motion.

Second, the simulation computer receives inputs from the angle of attack (α_m), pitch attitude motion variables (q_m and θ_m) observed in the wind tunnel test, and online solution result (γ_{corr}) of the aircraft's longitudinal trajectory. Based on the coupling relationship between the aircraft's translation and rotation, the online correction of the aircraft's pitch attitude motion is completed, and the corrected pitch angle θ_{corr} and pitch rate q_{corr} are obtained.

Finally, the online corrected pitch angle θ_{corr} and pitch rate q_{corr} are fed back to the flight control computer of the WTVFT, and the online correction test can be completed.

Fig. 6 depicts the WTVFT's flight control system. The test model has servos to drive deflections in the control surface, a flight control computer, rotary encoders, and an IMU. Angular velocities and three-axis attitude angles of the model are measured using an inertial measurement unit; the servos'

deflection angle is measured using rotary encoders. The flight control computer performs the control law algorithm, gathers information from the onboard sensors, and uses the servos to regulate the test model's attitude. Furthermore, the ground station is created and planned using the LabVIEW software framework, which can provide pertinent test data in real time, such as the servos' deflection angle, airflow angle, attitude angles, and angular velocities.

In the WTVFT, the control system software realizes network data communication between the main control computer and the flight control computer. The main control computer can realize data display, alternating between closed-loop and open-loop control, program and remote control switching; the flight control computer realizes data acquisition, recording, driving of the hardware and other functions.²¹

2.3. Online solution of data used for identification

Prior to identifying aerodynamic parameters using WTVFT data, the state of motion variables for identification needs to be solved online. The aerodynamic parameter identification is carried out using the test data obtained from the online correction test to obtain accurate aerodynamic derivatives.

2.3.1. Online solution for longitudinal trajectory and velocity

The longitudinal trajectory and velocity changes of the model can be solved online by the method of online correction test shown in Fig. 5. During the WTVFT, X_b , Y_b , and Z_b , the aerodynamic forces applied to the aircraft in the body coordinate system, can all be measured in real time by the force balance. Eq. (6) illustrates how the whole external force acting on the aircraft is mapped onto the airflow coordinate system.

$$\begin{cases} \begin{bmatrix} X_a \\ Y_a \\ Z_a \end{bmatrix} = \mathbf{L}_{ab} \begin{bmatrix} X_b + T \\ Y_b \\ Z_b \end{bmatrix} + \mathbf{L}_{ag} \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix} \\ \mathbf{L}_{ag} = \begin{bmatrix} \cos \gamma & 0 & -\sin \gamma \\ 0 & 1 & 0 \\ \sin \gamma & 0 & 1 \end{bmatrix} \\ \mathbf{L}_{ab} = \begin{bmatrix} \cos \alpha \cos \beta & \sin \beta & \sin \alpha \cos \beta \\ -\cos \alpha \sin \beta & \cos \beta & -\sin \alpha \sin \beta \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix} \end{cases} \quad (6)$$

where T is the thrust of the engine in equilibrium, $\mathbf{L}_{ab}$ is the transformation matrix from the body coordinate system to the airflow

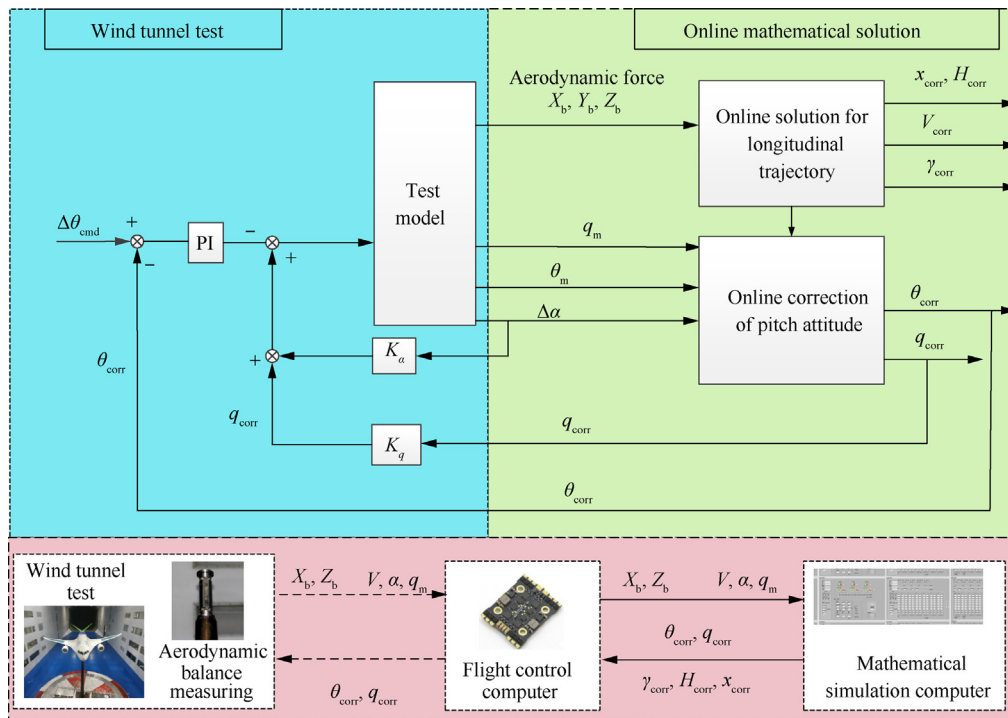


Fig. 5 Methodology for online correction test of longitudinal WTVFT.

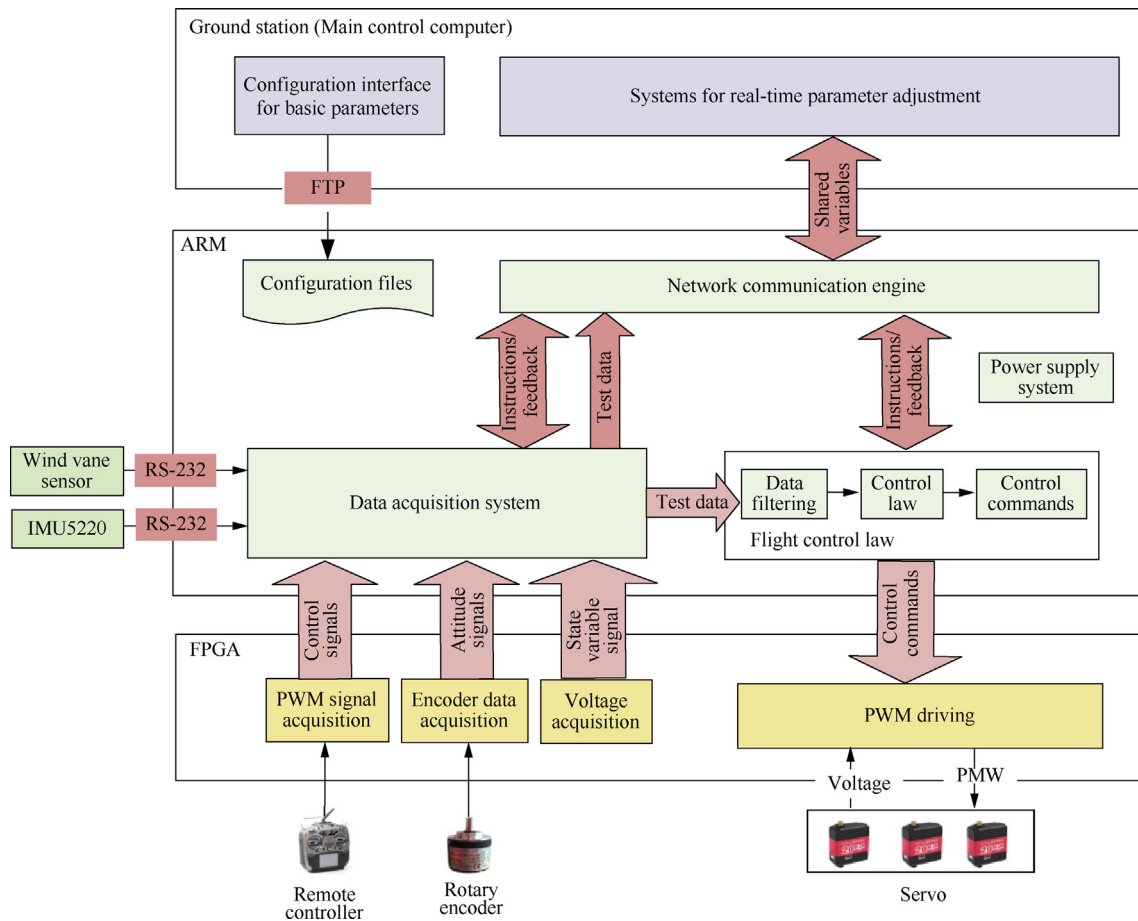


Fig. 6 Software scheme of WTVFT control system.

coordinate system, L_{ag} is the transformation matrix from the ground coordinate system to the airflow coordinate system, and X_a , Y_a , and Z_a are the aerodynamic forces applied to the model in the airflow coordinate system. Eq. (6) can be used to calculate the total external force operating on the airflow coordinate system's Z -axis, and the result is shown in Eq. (7):

$$Z_a = -(X_b + T) \sin \alpha + Z_b \cos \alpha + mg \cos \gamma \quad (7)$$

The variation rate of climb angle $\dot{\gamma}_{\text{corr}}$ for the test model can be obtained by the online mathematical solution of Eq. (8):

$$\dot{\gamma}_{\text{corr}} = \frac{-Z_a}{mV} = \frac{(X_b + T) \sin \alpha - Z_b \cos \alpha}{mV} - \frac{g}{V} \cos \gamma_{\text{corr}} \quad (8)$$

where γ_{corr} and $\dot{\gamma}_{\text{corr}}$ are the climb angle and the rate of change of climb angle obtained from the online solution, respectively. The climb angle γ_{corr} can be obtained by integrating the variation rate of the climb angle $\dot{\gamma}_{\text{corr}}$ obtained from the online mathematical solution. In addition, the velocity variation rate $\dot{V}_{\text{corr}}$ of the model can be solved online according to the aircraft centre-of-mass dynamics equations shown in Eq. (9).

$$\dot{V}_{\text{corr}} = \frac{T + X_b}{m} \cos \alpha + \frac{Z_b}{m} \sin \alpha - g \sin(\theta - \alpha) \quad (9)$$

where $\dot{V}_{\text{corr}}$ and V_{corr} are the variation rate of airspeed and airspeed obtained from the online solution, respectively. Integrating the rate change of airspeed $\dot{V}_{\text{corr}}$ obtained by the online mathematical solution can obtain the airspeed V_{corr} . According to the climb angle γ_{corr} and airspeed V_{corr} obtained by the online mathematical solution, the longitudinal trajectory of the aircraft can be solved online, as shown in Eq. (10):

$$\begin{cases} x_{\text{corr}} = \int_0^t V_{\text{corr}} \cos \gamma_{\text{corr}} \\ H_{\text{corr}} = \int_0^t V_{\text{corr}} \sin \gamma_{\text{corr}} \end{cases} \quad (10)$$

where x_{corr} and H_{corr} are the model's displacement in the horizontal direction and the model's height, respectively, as determined by the online solution.

2.3.2. Online corrections for pitching motion

Due to the coupling between the aircraft's longitudinal translational and pitching movements, there are differences between the pitching motion data obtained from the WTVFT and the free flight. As shown in Fig. 5, online corrections are also required for the variables θ and q of the pitching motion of the test model.

Fig. 7 illustrates how the pitch angle, angle of attack, and climb angle for free flight all meet the equation $\theta = \alpha + \gamma$ when lateral-directional motion is neglected. In contrast, during the WTVFT, the climb angle γ of the model is 0, thus, the angle of attack is about the same to the observed pitch angle $\theta_m = \alpha$. Furthermore, the pitch rate determined by the test is equivalent to the rate of change of the model's angle of attack since the pitch rate is roughly equal to the derivative of the pitch angle, i.e., $q_m = \dot{\alpha}$.

As shown in Fig. 5, the online correction of pitch motion data is carried out by the online correction test of the WTVFT, and the pitch rate q_m recorded by the sensor in the wind tunnel test is able to be added to the change rate of the climb angle $\dot{\gamma}_{\text{corr}}$ obtained from the online solution of Eq. (8) to produce the online corrected pitch rate q_{corr} , as shown in Eq. (11). Integrating the corrected pitch rate q_{corr} gives the corrected pitch angle θ_{corr} .

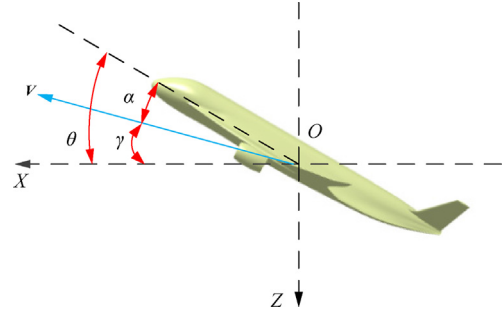


Fig. 7 Coupling of longitudinal translational motion and rotational motion.

$$\begin{cases} q_{\text{corr}} = q_m + \dot{\gamma}_{\text{corr}} \\ \theta_{\text{corr}} = \int_0^t q_{\text{corr}} \end{cases} \quad (11)$$

The online corrected pitch angle θ_{corr} and pitch rate q_{corr} are then input to the flight control law computer of the WTVFT to complete the correction of pitch motion. In summary, the data required for identification obtained after the online correction test and data solving for wind tunnel virtual flight are α_{corr} , x_{corr} , H_{corr} , γ_{corr} , V_{corr} , θ_{corr} , and q_{corr} .

3. Aerodynamic parameter identification method based on online correction test data

3.1. Identification model based on corrected data

According to the analysis in Section 2.1, the trajectory and airspeed of the model do not change during the conventional uncorrected WTVFT, it is thus impossible to identify the test model's normal and axial force derivatives directly using the equations of motion shown in Eq. (1). In this case, using a force balance to measure the aerodynamic forces acting on the entire aircraft throughout real time and using the observed values of the axial force X and normal force Z as observation variables for identifying the aerodynamic force derivatives are required.¹⁰

The corrected WTVFT data contain the longitudinal trajectory of the model as well as the velocity changes. Therefore, for the corrected WTVFT data, it is possible to identify aerodynamic parameters by using the unconstrained free flight equations of motion.

There are three components to the WTVFT's longitudinal identification model: aerodynamic model, state equation and observation equation. Eq. (12) displays the aircraft's longitudinal aerodynamic model for small and medium angles of attack.

$$\begin{cases} M = \bar{q} S c [C_{m*} + C_{m\alpha}(\alpha - \alpha_*) + C_{mq}qc/2V + C_{m\dot{\alpha}}\dot{\alpha}c/2V + C_{m\delta e}\delta_e] \\ Z_b = \bar{q} S [C_{Z*} + C_{Z\alpha}(\alpha - \alpha_*) + C_{Zq}qc/2V + C_{Z\delta e}\delta_e] \\ X_b = \bar{q} S [C_{X*} + C_{X\alpha}(\alpha - \alpha_*) + C_{Xq}qc/2V + C_{X\delta e}\delta_e] \end{cases} \quad (12)$$

where $\bar{q}$ is the dynamic pressure, S is the wing area, and c is the mean aerodynamic chord, α_* is the angle of attack during the trimmed flight state, and the aircraft's pitching moment, normal force, and axial force coefficients in trimmed flight are represented by the values C_{m*} , C_{Z*} , and C_{X*} , respectively. The parameters to be identified are $\Theta = [C_{m*}, C_{m\alpha}, C_{mq}, C_{m\dot{\alpha}}, C_{m\delta e}, C_{Z*}, C_{Z\alpha}, C_{Zq}, C_{Z\delta e}, C_{X*}, C_{X\alpha}, C_{Xq}, C_{X\delta e}]$.

For the corrected test data of the WTVFT, the state equations for the longitudinal aerodynamic parameter identification are shown in Eq. (13):

$$\begin{cases} \dot{V}_{\text{corr}} = \frac{T+X_b}{m} \cos \alpha + \frac{Z_b}{m} \sin \alpha - g \sin(\theta_{\text{corr}} - \alpha) \\ \dot{\alpha}_{\text{corr}} = q_{\text{corr}} + \frac{g}{V_{\text{corr}}} \cos(\theta_{\text{corr}} - \alpha) - \frac{T+X_b}{mV_{\text{corr}}} \sin \alpha + \frac{Z_b}{mV_{\text{corr}}} \cos \alpha \\ \dot{\theta}_{\text{corr}} = q_{\text{corr}} \\ \dot{q}_{\text{corr}} = \frac{M}{I_y} \\ \dot{H}_{\text{corr}} = V_{\text{corr}} \sin(\theta_{\text{corr}} - \alpha) \end{cases} \quad (13)$$

The observation variables are the test data $V_{\text{corr-m}}$, $\alpha_{\text{corr-m}}$, $H_{\text{corr-m}}$, $\theta_{\text{corr-m}}$, and $q_{\text{corr-m}}$ that were acquired after the online correction test and data solution; the observation equations can be seen in Eq. (14):

$$X_{\text{corr-m}} = X_{\text{corr}} + v_X \quad (14)$$

where X_{corr} is the theoretical value of the test data corrected for WTVFT, v_X is the measurement noise, and X is the observation variables V , α , θ , q , and H .

3.2. Design methods for excitation signals and identification algorithms

Excitation signals such as sine, frequency sweep, dipole square wave, and 3211 multipole square wave are often used for identifying aerodynamic parameters.²⁵ When an aircraft employs a flight control law, the oscillatory motion is suppressed by the flight control law. 3211 multipole square wave and dipole square wave signals show significant cancellation and attenuation when they are input to the aircraft's control surfaces. In Ref. 26, that is suggested to choose the frequency sweep signal as the excitation signal for aerodynamic parameter identification of the aircraft equipped with a control augmentation system, as it is less affected by the flight control law than other excitation signals.

As shown in Fig. 8, a frequency domain analysis-based approach is presented in this research to design the excitation signal's frequency range and amplitude magnitude. By analysing the model (Eq. (13)) established in Section 3.1 for identification in the frequency domain, each parameter that has to be identified is monitored in relation to the frequency fluctuation of the excitation signal in terms of its amplitude-frequency response. To find the excitation signal's frequency range, make sure the amplitude response of the aerodynamic forces or moments brought about by each motion variable and manipulation variable is of the same order of magnitude. Ultimately, the excitation signal's amplitude is adjusted to achieve that the aircraft's motion variables respond significantly within the intended frequency range.

The parameters that need to be determined for the frequency sweep signal are the lower frequency limit ω_l , the upper frequency limit ω_h , and the amplitude of the signal $|A|$. The equations for the normal force, axial force, and pitching moment are included in Eq. (15), which is the observation equation for the test model's longitudinal motion.

$$\begin{cases} \Delta Z_b = (C_{Z\alpha}\Delta\alpha + C_{Zq}\Delta q + C_{Z\delta e}\Delta\delta_e) \cdot \bar{q} S \\ \Delta X_b = (C_{X\alpha}\Delta\alpha + C_{Xq}\Delta q + C_{X\delta e}\Delta\delta_e) \cdot \bar{q} S \\ \Delta \dot{q} = [C_{m\alpha}\Delta\alpha + (C_{mq} + C_{m\dot{\alpha}})\Delta q + C_{m\delta e}\Delta\delta_e] \cdot \bar{q} S c / I_y \end{cases} \quad (15)$$

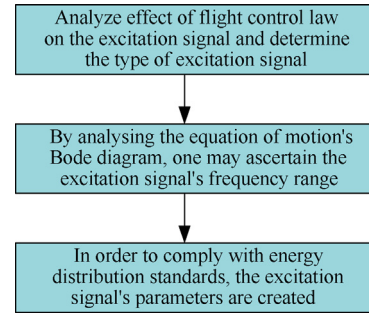


Fig. 8 Process of longitudinal excitation signal design for WTVFT.

where ΔX_b , ΔZ_b , $\Delta \dot{q}$ are the changes in axial force, normal force, and pitch angle angular acceleration of the model measured during the WTVFT, respectively. To see the amplitude-frequency responses of the normal force, axial force, and pitching moment components shown in Eq. (15) with the change in the excitation signal's frequency, the Bode diagram analysis of the identification model utilised for the corrected test data shown in Eq. (13) is carried out.

The pitching moment equation is used to illustrate the amplitude-frequency response curves of the three pitching moment components as the elevator signal's frequency changes: $|C_{m\alpha}\Delta\alpha/\delta_e|$, $|(C_{mq} + C_{m\dot{\alpha}})\Delta q/\delta_e|$, and $|C_{m\delta e}\Delta\delta_e/\delta_e|$. In accordance with the normal and axial force observation equations, the amplitude-frequency response curves of the longitudinal aerodynamic components along the frequency variation in the elevator excitation signals are shown: $|C_{Z\alpha}\Delta\alpha/\delta_e|$, $|C_{Z\delta e}\Delta\delta_e/\delta_e|$, $|C_{Zq}\Delta q/\delta_e|$, $|C_{X\alpha}\Delta\alpha/\delta_e|$, $|C_{X\delta e}\Delta\delta_e/\delta_e|$, and $|C_{Xq}\Delta q/\delta_e|$.

Fig. 9 displays the amplitude-frequency characteristic curves corresponding to every pitching moment component while the excitation signal's frequency changes. The aircraft's short-period modal frequency is nearly equivalent to the frequencies corresponding to the maxima of the curves in Fig. 9. Additionally, the response amplitudes of all pitching moment components drop as the frequencies of the excitation signals go beyond a particular location. The same method can be used to produce the amplitude-frequency characteristic

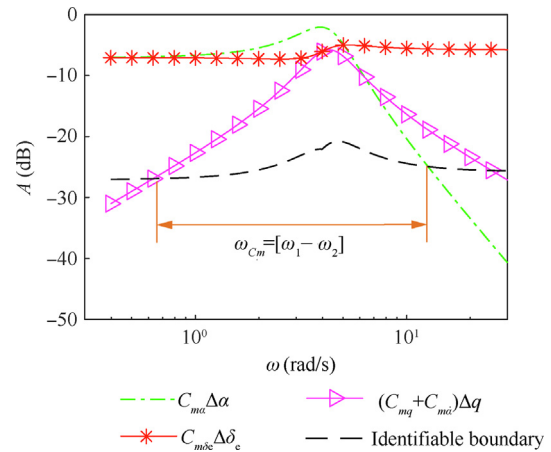


Fig. 9 Amplitude-frequency characteristics of pitching moment components with frequency of excitation signals.

curves of the axial force and normal force components with a variation in the excitation signal's frequency.

The pitching moment components' magnitude responses $C_{mz}\Delta\alpha$, $(C_{mq} + C_{m\dot{z}})\Delta q$, and $C_{m\delta e}\Delta\delta_e$ resulting from variations in the servo's deflection, pitch rate, and angle of attack should not differ much in order to guarantee the accuracy of each derivative's identification. In the same way, for the normal force and axial force observation equations shown in Eq. (15), the magnitude responses of each normal force component $C_{Zz}\Delta\alpha$, $C_{Zq}\Delta q$, $C_{Z\delta e}\Delta\delta_e$ and axial force component $C_{Xz}\Delta\alpha$, $C_{Xq}\Delta q$, $C_{X\delta e}\Delta\delta_e$ ought should be a bit comparable.

Fig. 10 shows the time-domain variation curves of each pitching moment component at various elevator excitation signal input frequencies. Pitch rate q in Fig. 10(a) approximates to 0 when the elevator excitation signal frequency is too low. The longitudinal static stability moment $C_{mz}\Delta\alpha$ and the manoeuvring moment $C_{m\delta e}\Delta\delta_e$ approximated to cancel each other, which results in the final full-aircraft pitching moment being small and thus not able to excite a significant pitch rate,

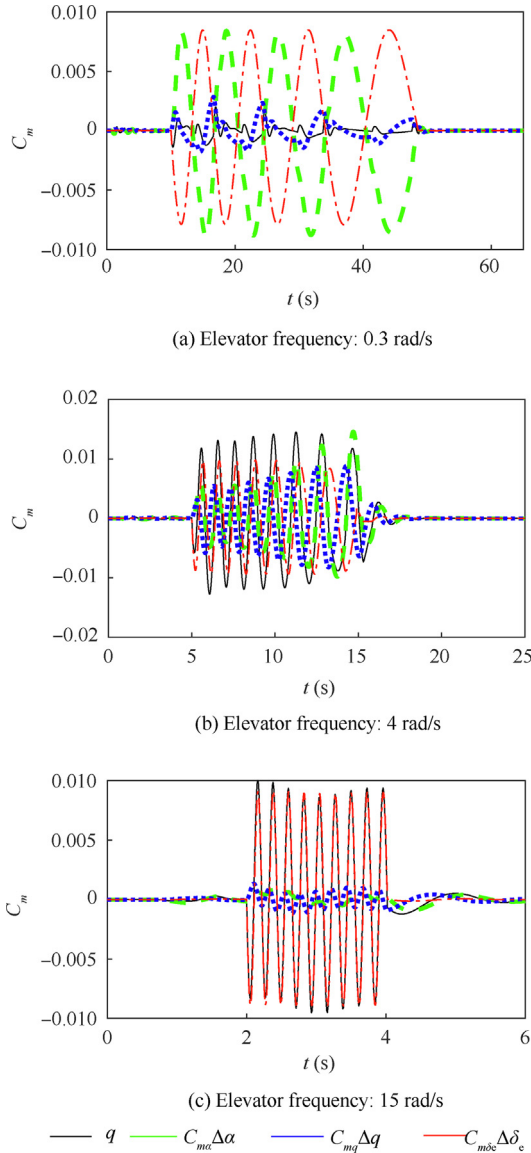


Fig. 10 Pitching moment components' time-domain response to various frequency excitation signal.

and thus, the pitching moment derivatives cannot be precisely identified. In Fig. 10(c), it can be shown that if the frequency of the elevator excitation signal is excessively high, the reaction of the aircraft's pitch is much slower compared to the elevator's manoeuvre speed. Consequently, this leads to a little alteration in the longitudinal motion variables (α , q) of the aircraft, so they are unable to be identified precisely. In Fig. 10(b), the identification results are accurate only if the amplitude responses of all pitch motion variables are on an order of magnitude when the frequency of the elevator excitation signal is within the appropriate range.

The aerodynamic derivative corresponding to a component is deemed recognisable when its contribution to the overall aerodynamic force or moment of the aircraft is at least 10%, as mentioned in Ref. 27. In order to do this, the Bode diagram's response amplitudes for each aerodynamic force and moment component in a certain frequency range are added together, and the value equal to one-tenth of the overall response amplitude is determined, as demonstrated in Eq. (16):

$$\begin{cases} |A|_{Cm-\omega_i} = \frac{1}{10} \sum \omega_i \left(|A|_{Cmz\Delta\alpha} + |A|_{(Cmq+Cm\dot{z})\Delta q} + |A|_{Cm\delta e\Delta\delta_e} \right) \\ |A|_{CZ-\omega_i} = \frac{1}{10} \sum \omega_i \left(|A|_{CZz\Delta\alpha} + |A|_{CZq\Delta q} + |A|_{CZ\delta e\Delta\delta_e} \right) \\ |A|_{CX-\omega_i} = \frac{1}{10} \sum \omega_i \left(|A|_{CXz\Delta\alpha} + |A|_{CXq\Delta q} + |A|_{CX\delta e\Delta\delta_e} \right) \end{cases} \quad (16)$$

where $|A|_{Cmz\Delta\alpha}$, $|A|_{(Cmq+Cm\dot{z})\Delta q}$, $|A|_{Cm\delta e\Delta\delta_e}$ are the pitching moment components' response amplitudes at the excitation signal frequency ω_i ; $|A|_{CZz\Delta\alpha}$, $|A|_{CZq\Delta q}$, $|A|_{CZ\delta e\Delta\delta_e}$ are the response amplitudes of the normal force components; and $|A|_{CXz\Delta\alpha}$, $|A|_{CXq\Delta q}$, $|A|_{CX\delta e\Delta\delta_e}$ are the response amplitudes of the axial force components. The values of $|A|_{Cm-\omega_i}$, $|A|_{CZ-\omega_i}$, and $|A|_{CX-\omega_i}$ represent one-tenth of the total response amplitudes of the axial force, normal force, and pitching moment for the excitation signal's frequency ω_i .

As per Eq. (16), the dividing line in Fig. 9 that represents one-tenth of the overall pitching moment response amplitude is shown by the dashed line, designating it as the identifiable boundary of each pitching moment derivative. In the same way, the identifiable boundaries of the derivatives of the axial force and normal force can be derived.

The aerodynamic derivative is recognisable at this frequency if and only if the response amplitude of the force or moment that corresponds to that aerodynamic derivative exceeds the identifiable boundary. Every pitching moment derivatives are recognised when the excitation signal frequency falls between $\omega_{Cm} = [\omega_1-\omega_2]$ in Fig. 9, meaning that all pitching moment components' amplitude responses are beyond the identifiable limit. Consequently, the excitation signal's frequency range, as established by the pitching moment derivatives, is ω_{Cm} .

By the same token, the frequency bands of the excitation signal determined by the normal force and axial force derivatives can be obtained in the range of ω_{CZ} and ω_{CX} . To satisfy the requirement of identifiability of all aerodynamic derivatives in the longitudinal direction, it is necessary to solve for the intersection of ω_{Cm} , ω_{CZ} , and ω_{CX} to obtain the frequency band of the longitudinal excitation signal:

$$\omega = \omega_{Cm} \cap \omega_{CZ} \cap \omega_{CX} \quad (17)$$

There are many methods used for aerodynamic parameter identification.²⁸ The output error method, which is based on maximum likelihood estimation, is the most often used technique for parameter identification of the nonlinear model described in Eqs. (12)–(14).²⁹ In order to determine the parameter to be identified, the maximum likelihood estimation involves creating a likelihood function with the observations and the unknown parameter as the independent variables. The extreme value of the likelihood function is then found by solving for it.

4. Validation of parameter identification method based on online correction test

Through the discussion in the previous two sections, a method for longitudinal aerodynamic parameter identification based on the online correction test of WTVFT has been developed, and Fig. 11 depicts the particular steps. The flight dynamics equations of the test model's WTVFT are first constructed in order to analyse the differences between the longitudinal dynamics characteristics of the WTVFT and the free flight, as well as to determine the pitch motion variables (θ and q) that require online correction and the trajectory variables (γ , H , and V) that require online solution. Second, an identification model applicable to the corrected data is established. The identification model's Bode diagram is analysed to ascertain the excitation signal's frequency range and amplitude magnitude, taking into account the recognisable requirements of the aerodynamic derivatives. Furthermore, the aerodynamic forces obtained from the online correction test are used to solve the model's longitudinal trajectory and velocity changes online. Pitch angle and pitch rate adjustments are also carried out online at the same time. Finally, the aerodynamic derivatives are determined, and the corrected WTVFT data is used to confirm the identification results.

As shown in Fig. 12, the online correction test based on WTVFT was completed in an FL-10 low-speed wind tunnel at the Aerodynamic Research Institute of China Aviation Industry Corporation (AVIC). The test part of the wind tunnel is 20 m long, 8 m wide, and has a maximum wind speed of 110 m/s.³⁰

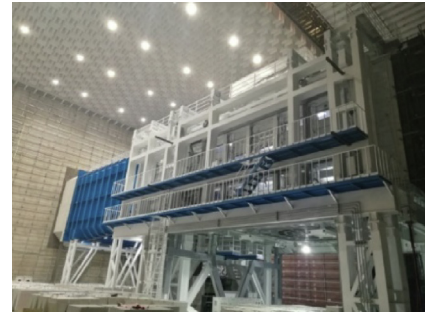


Fig. 12 Low-speed wind tunnel (FL-10).

The test model comprises a blended wing body and V-tail configuration, as seen in Fig. 13. The horizontal and vertical tails are replaced with a V-tail, which functions as an elevator δ_e when pitch motion because the servos on each side of the V-tail are deflected in the same direction.³¹ The test model's scaling ratio is $k = 1/9$, and Table 3 illustrates how the particular test model parameters can be determined by applying a similarity criteria to the scaled model and the full-size aircraft body characteristics.^{32,33}

Pitch angle command drives the test model's longitudinal flight control rule, as Fig. 5 illustrates. The longitudinal flight control law parameters are $K_{p\theta} = 1.34$, $K_q = 0.82$, $K_\alpha = 1.5$, and $K_{I\theta} = 0.7$.

4.1. Excitation signal's design

In accordance with the force measurements obtained through the conventional wind tunnel test displayed in Fig. 14, the model of longitudinal motion presented in Eq. (13) applicable to the corrected test data is produced. Fig. 15 displays the plotted amplitude-frequency responses of the pitching moment, normal force, and axial force components with the elevator excitation signal's frequency change. The short-period modal frequency of the longitudinal direction of the test model is 4 rad/s, and the peaks of the curves in Fig. 15 are near the short-period modal frequency.

Fig. 15 shows the identifiable boundaries of the axial force, pitching moment, and normal force derivatives, indicated by

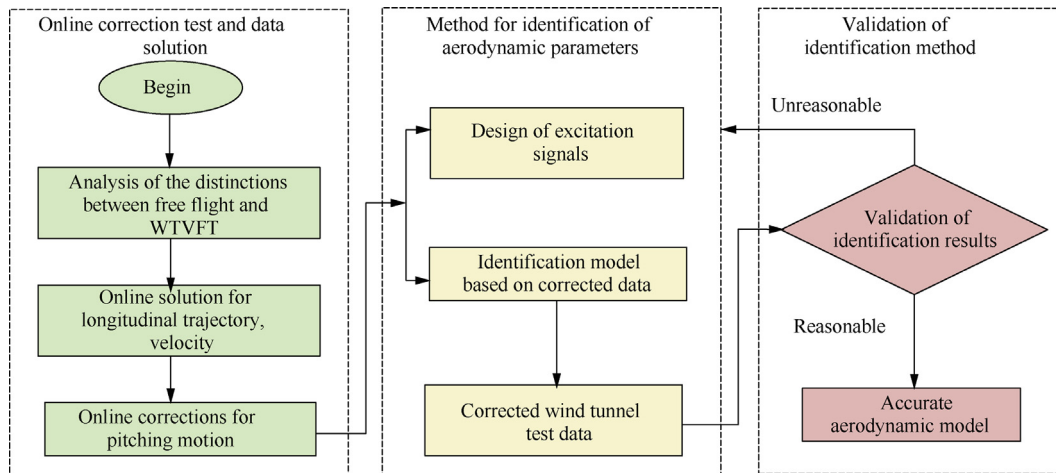


Fig. 11 Procedures for identifying aerodynamic parameters for an online corrective test using WTVFT.

dashed lines, based on the design procedure for the excitation signals described in Section 3.2. The amplitude responses of all force and moment components must be beyond the identifiable boundary in order for each aerodynamic derivative to be identifiable. Accordingly, the excitation signals' frequency band ranges, which are derived from the derivatives of the pitching moment, normal force, and axial force, are $\omega_{Cm} = [0.7, 10.5]$ rad/s, $\omega_{CZ} = [2.5, 25]$ rad/s, and $\omega_{CX} = [2.3, 25]$ rad/s. To get the longitudinal excitation signal's frequency range, consider the intersection of ω_{Cm} , ω_{CZ} , and ω_{CX} :

$$\begin{aligned}\omega &= \omega_{Cm} \cap \omega_{CZ} \cap \omega_{CX} \\ &= [0.7, 10.5] \cap [2.5, 25] \cap [2.3, 25] \\ &= [2.5, 10.5] \text{ rad/s}\end{aligned}\quad (18)$$

In conclusion, in order to guarantee the longitudinal aerodynamic parameters of the model can be identified, the frequency range of the frequency sweep excitation signal has to fall within $[2.5, 10.5]$ rad/s, roughly corresponding to the frequency range of 0.5 to 2.5 times the short-period mode. Considering the limitation of the deflection rate of the servos, the frequency of the sweep signal is chosen as $\omega = [2.5, 7]$ rad/s in this experiment. The amplitude of different excitation signals is selected, and through mathematical simulation, the change range for the angle of attack of the test model is within $\pm 2^\circ$, and the amplitude of the excitation signal for this test is finally determined to be $|A| = 2.8^\circ$, as shown in Fig. 16.

4.2. Analysis of data from online correction test

To realize the online correction test for the WTVFT and the solution of the identification data shown in Fig. 5, the longitudinal trajectory and velocity change of the model are mathematically solved online using the translational equations of motion for the model, and the aerodynamic force supplied to the test model is recorded in real time using the strain gauge

Table 3 Ontological parameters of the model.

Parameter	Proportion	Full-size aircraft	Test model
Wing span b (m)	1/9	36	4.00
Mean aerodynamic chord c (m)	1/9	10.41	1.16
Wing area S (m ²)	(1/9) ²	241	2.98
Mass m (kg)	(1/9) ³	49149	67.42
Pitching moment of inertia I_y (kg·m ²)	(1/9) ⁵	4044856	68.50
Yawing moment of inertia I_z (kg·m ²)	(1/9) ⁵	5166787	87.50
Rolling moment of inertia I_x (kg·m ²)	(1/9) ⁵	1210504	20.50
Product of inertia I_{xz} (kg·m ²)	(1/9) ⁵	171242	2.90

balance throughout the test. Simultaneously, the online pitch motion data of the model are rectified, and the corrected pitch angle and pitch rate are sent into the test model's flight control law to regulate the model's motion in the wind tunnel, as shown in Fig. 17.

The longitudinal frequency sweep excitation signal designed in Section 4.1 (shown in Fig. 16) is input to the pitch control surfaces of the test model, and the data from the online correction test as well as the data from the conventional 3-DOF WTVFT are recorded separately. Meanwhile, the longitudinal motion simulation model of unconstrained free flight is established based on the aerodynamic data obtained from the conventional wind tunnel force measurement test seen in Fig. 14. Under the same excitation signal input, the free flight simulation data are compared with the WTVFT test data, as seen in Fig. 18.

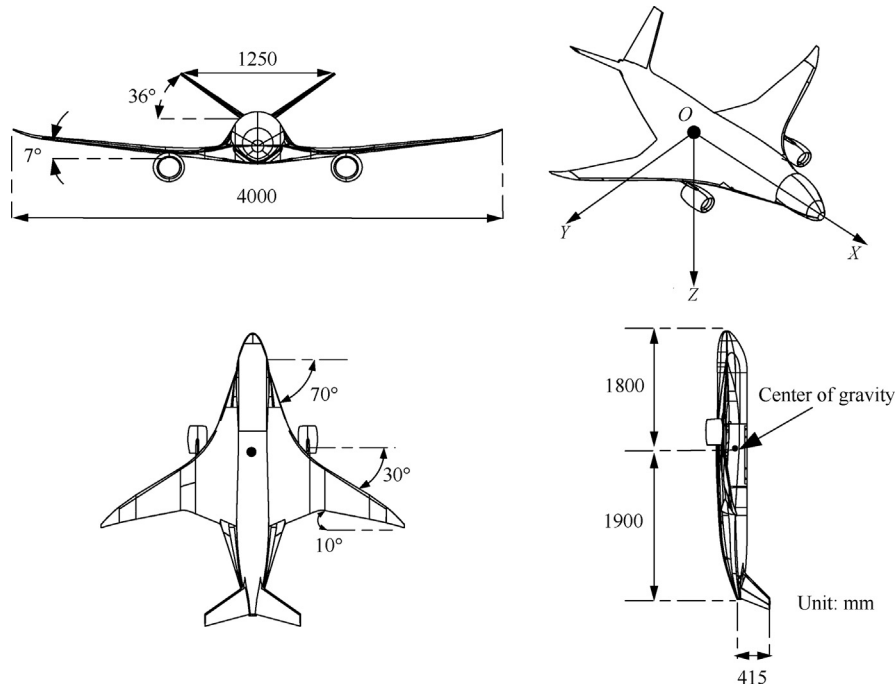


Fig. 13 General layout parameters of the model.

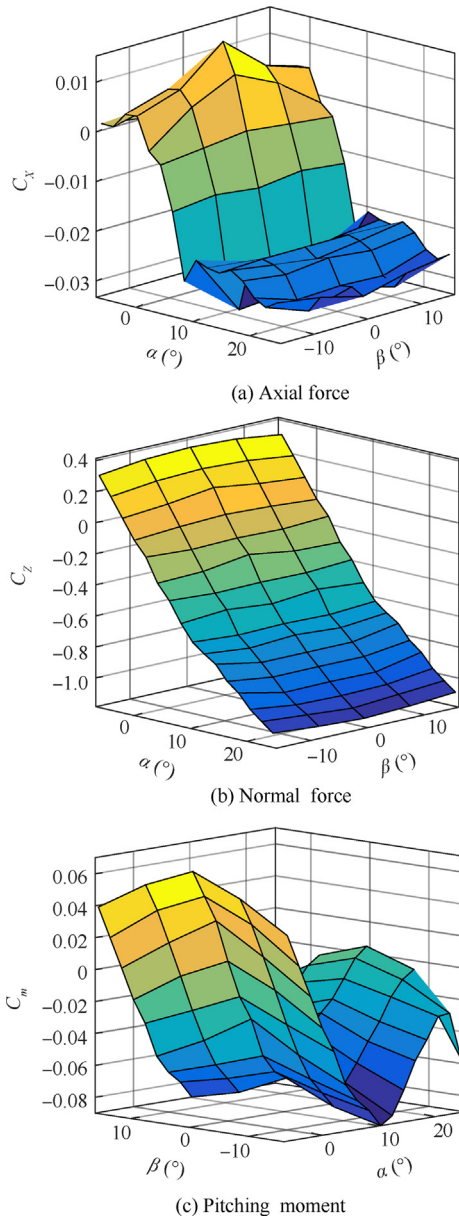


Fig. 14 Data from conventional wind tunnel force measurements of test model ($V = 30$ m/s).

In addition, the Goodness-of-Fit (GOF) of the two WTVFT data and the free flight simulation data are shown in Table 4. The mutual fit of the two sets of data is measured by the GOF, and the closer the GOF is to 1, the better the fit between the two sets of data.³⁴

According to Fig. 18, the responses of airspeed, climb angle, and altitude obtained from the online correction test have small deviations from the unconstrained free flight motion simulation data, and the GOFs are above 0.9. The absence of longitudinal trajectory and velocity changes in the conventional uncorrected WTVFT causes a large discrepancy between the test data and the free flight simulation results, and thus, the GOFs of the airspeed, climb angle, and altitude responses are lower than 0.3. In the online correction test, the longitudinal trajectory and velocity of the model are obtained

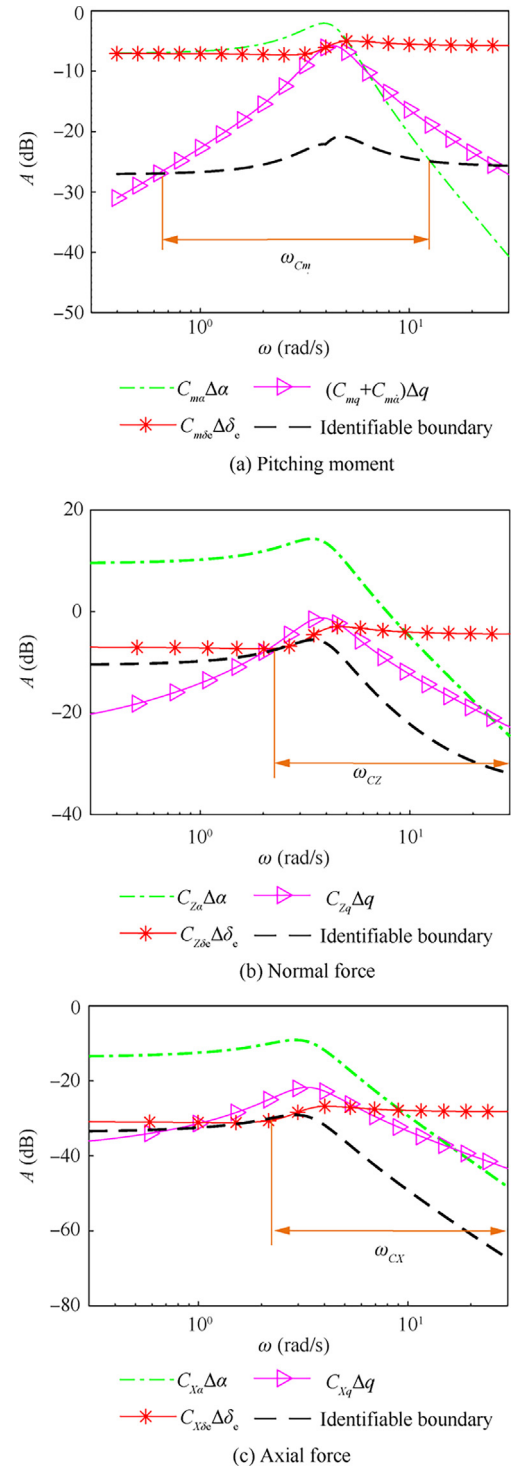


Fig. 15 Amplitude-frequency characteristics of longitudinal aerodynamic parameters as a function of excitation signal frequency.

from the online mathematical solution of the aircraft's centre-of-mass translational equations of motion, and thus, the longitudinal trajectory and velocity data are well fitted to free flight.

In both two wind tunnel tests, the test model has pitch motion, and the pitch angle and pitch rate responses of the corrected tests fit the free flight simulation data to a high degree,

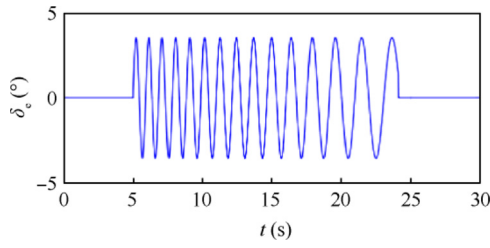


Fig. 16 Identification of excitation signal in longitudinal direction.



Fig. 17 Installation of the model for 3-DOF WTVFT.

with GOFs higher than 0.95, while the GOFs of the pitch motion in the conventional uncorrected tests are approximately 0.85. According to Eq. (5), the translational and rotational motions of the model are coupled with each other, and the model in the conventional uncorrected test does not have translational motion, which causes it to differ from the pitch motion of free flight.

In summary, the corrected WTVFT data are approximately consistent with the unconstrained free flight simulation results, but the conventional uncorrected WTVFT data have a large difference from the unconstrained free flight simulation results.

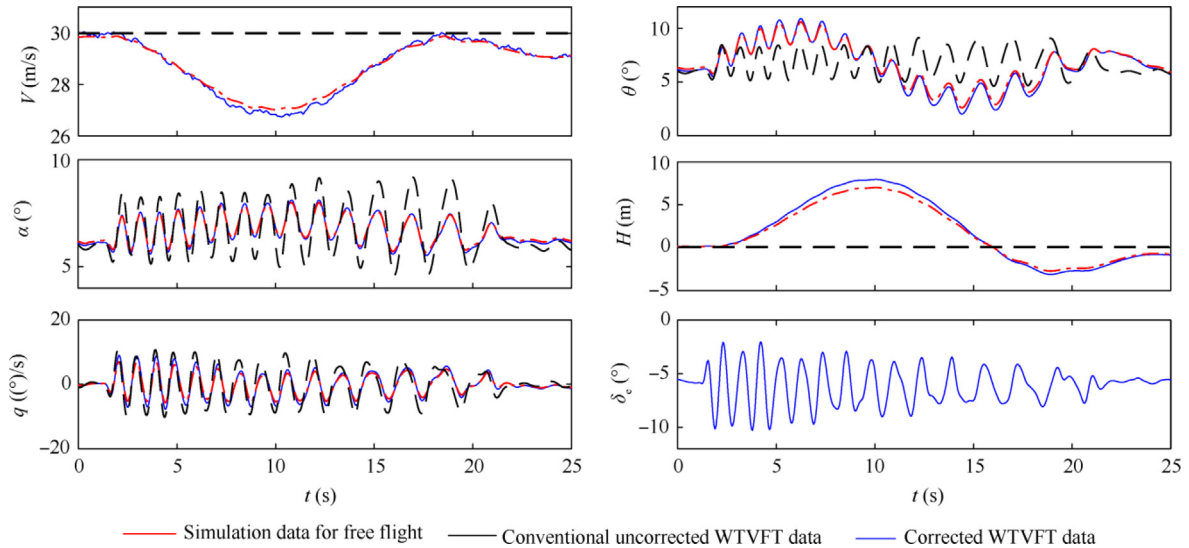


Fig. 18 Comparative analysis of online correction test results for wind tunnel virtual flights.

Table 4 GOF of two WTVFT data to free flight simulation data.

Parameters	Conventional uncorrected test data	Corrected test data
θ (°)	0.866	0.951
V (m/s)	0.282	0.969
α (°)	0.253	0.935
q (°/s)	0.851	0.956
H (m)	0.178	0.923

4.3. Results and valuation of aerodynamic parameter identification

The conventional uncorrected test data V_m , α_m , θ_m , q_m , X_{b-m} , and Z_{b-m} and the corrected test data θ_{corr} , q_{corr} , α_m , x_{corr} , H_{corr} , γ_{corr} , and V_{corr} are obtained through the WTVFT carried out in Section 4.2.

The two sets of WTVFT illustrate the maximum likelihood technique used to determine the longitudinal aerodynamic derivatives. Table 5 displays the results of the identification, and the deviation of the identification results from the conventional wind tunnel force measurement results is calculated.

Notably, in the conventional uncorrected WTVFT, the combined external force of the model is 0. It is necessary to measure the aerodynamic force of the whole aircraft in real time by means of a force balance and identify the aerodynamic force derivatives by considering the measured values of the normal force Z and the axial force X as observation variables. In contrast, in the online correction test, the aerodynamic derivatives are obtained by identifying the corrected test data, as shown in Eq. (13).

- (1) It is evident from a comparison of Table 5's identification results for the axial and normal force derivatives that the two test data's identification results mostly differ by 10% from the conventional wind tunnel force measurement results.

Table 5 Identification results of aerodynamic parameters.

Parameter		Measurements from conventional wind tunnel test	Conventional uncorrected test data		Online corrected test data	
			Identification result	Deviation (%)	Identification result	Deviation (%)
Normal & axial force derivatives	C_{Z^*}	-0.022	-0.023	4.50	-0.024	9.10
	$C_{Z\alpha}$	-3.99	-3.80	4.80	-3.78	5.30
	C_{Zq}	-12.12	-12.58	3.80	-12.86	6.10
	$C_{Z\delta e}$	-0.19	-0.17	10.50	-0.17	10.50
	C_{X^*}	-0.003	-0.002	33.3	-0.002	33.30
	$C_{X\alpha}$	0.33	-0.023	3.0	0.34	-3.0
	C_{Xq}	-0.95	-3.80	3.20	-1.03	8.40
	$C_{X\delta e}$	-0.03	-12.58	30.0	-0.023	23.30
	C_{m^*}	-0.059	-0.065	10.20	-0.054	8.50
Pitching moment derivatives	$C_{m\dot{z}}$	-0.52	-0.69	32.70	-0.50	3.80
	$C_{mq} + C_{m\dot{x}}$	-5.68	-5.76	1.40	-5.72	0.70
	$C_{m\delta e}$	-0.51	-0.57	11.70	-0.49	3.90

- (2) By comparing the identification results of the pitching moment derivatives in Table 5, it can be seen that the deviation of the longitudinal stability derivative $C_{m\dot{z}}$ in the conventional uncorrected test is greater than 30%, and the deviation of the manoeuvring derivative $C_{m\delta e}$ is greater than 10%, whereas in the corrected test, the deviation of both is less than 5%.

According to the analysis in Section 4.2, the translational and rotational motions of the aircraft are coupled with each other, and the absence of translational motion of the model in the conventional uncorrected WTVFT leads to a GOF with unconstrained free flight pitch motion of less than 0.9. As a result of this discrepancy in the dynamics, the accuracy of the longitudinal moment derivatives $C_{m\dot{z}}$ and $C_{m\delta e}$ obtained from the identification is degraded. The pitch motion data of the corrected WTVFT are close to the results of the free flight, and the GOF of its test data and the free flight simulation data are all greater than 0.95, so the deviation of the identification results of the pitching moment derivatives is small.

In summary, the aerodynamic derivatives identified by the online corrected test are more accurate than the identification results of the conventional uncorrected test data.

5. Conclusions

- (1) For the first time, the online correction test and data solving method for WTVFT are proposed and realized. It is obtained from the analysis that displacement constraints and velocity invariance are the main reasons for the difference in longitudinal dynamic characteristics between the WTVFT and free flight of the aircraft. The longitudinal trajectory and velocity changes of the model are solved online based on the translational equations of motion of the aircraft, and the online correction of the pitching motion data is completed based on the coupling relationship between the translational and rotational motion of the aircraft, which eliminates the difference in the flight dynamics between the WTVFT and the free flight.

- (2) The longitudinal trajectory, velocity, and pitch motion variables obtained by solving the data from the online correction test are used as state and observation variables to establish the identification model based on the corrected test data. In addition, a design method of excitation signal parameters based on frequency domain analysis is innovatively established. The excitation signal frequency is approximately within 0.5–2.5 times the short-period modal frequency, which ensures the identifiability of each aerodynamic derivative.
- (3) Compared with the conventional uncorrected WTVFT, the test data obtained from the online correction test are more consistent with the simulation results of unconstrained free flight. Moreover, the aerodynamic derivatives obtained from the identification based on the online correction test data are more accurate than the results obtained from the identification based on the conventional WTVFT.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Video: For a video summary of this paper, please visit the supplementary material. Supplementary data to this paper can be found online at <https://doi.org/10.1016/j.cja.2024.05.031>.

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